

## ASSIGNMENT - 1

192311159

## UNIT - 1

- 1) a) Encrypt the message "DEFEND THE WALL" using caesar cipher with Key = 3.
- b) Decrypt the message "KHOO RZRUDG" using caesar cipher with Key = 3.
- c) A caesar cipher is used to encrypt the word "SECRET" to "VHFUHW". Determine the key used and decrypt "WKLV LV D WHVW".

2) a) CT: G H I H A G W K H Z D O O

| PT        | D | E | F | E | N  | O | T  | H  | E | W  | A | L  | L  |
|-----------|---|---|---|---|----|---|----|----|---|----|---|----|----|
| VAL       | 3 | 4 | 5 | 4 | 13 | 3 | 19 | 7  | 4 | 22 | 0 | 11 | 11 |
| Key       | 3 | 3 | 3 | 3 | 3  | 3 | 3  | 3  | 3 | 3  | 3 | 3  | 3  |
| PT+K      | 6 | 7 | 8 | 7 | 16 | 6 | 22 | 10 | 7 | 25 | 3 | 14 | 14 |
| %26       | 6 | 7 | 8 | 7 | 16 | 6 | 22 | 10 | 7 | 25 | 3 | 14 | 14 |
| <u>CT</u> | G | H | I | H | A  | G | W  | K  | H | Z  | D | O  | O  |

100  
100  
100

b)

| CT   | K  | H | O  | O  | R  | Z  | R  | U  | O  | G |
|------|----|---|----|----|----|----|----|----|----|---|
| VAL  | 10 | 7 | 14 | 14 | 17 | 25 | 17 | 20 | 14 | 6 |
| Key  | 3  | 3 | 3  | 3  | 3  | 3  | 3  | 3  | 3  | 3 |
| CT-K | 7  | 4 | 11 | 11 | 14 | 22 | 14 | 17 | 11 | 3 |
| %26  | 7  | 4 | 11 | 11 | 14 | 22 | 14 | 17 | 11 | 3 |
| PT   | H  | E | L  | L  | O  | W  | O  | R  | L  | D |

PT: HELLO WORLD

c) SECRET → VHFUHW

each letter is shifted by +3, Key = 3.

Decrypt: WKLV LV D WHVW

P.T: THIS IS A TEST

2) a) using the substitution key:

A → Q, B → W, C → E, D → R, E → T, F → Y, ..., Z → H

Encrypt the word "FIRE".

b) Given this cipher text "UJHNO QEB NRFZH YOLTK" and knowing it is encrypted with monoalphabetic substitution Find the plain text.



a) monoalphabetic

A → Q, B → W, C → E, D → R, E → T, F → Y, G → U, H → I, I → O, J → P, K → S,  
L → H, M → D, N → F, O → G, P → M, Q → J, R → K, S → L, T → Z, U → X,  
V → C, W → V, X → B, Y → N, Z → M

Encrypt: FIRE

F → Y

I → O

R → K

E → T

Cipher text: YOKT

b) Frequency Analysis

Cipher Text: UJHHD AEB NRFZH YOLTK

Best known plaintext: HELLO THE WORLD AGAIN

3) a) Encrypt the message "ATTACK AT PLACE" using the key word "LEMON".

b) Decrypt the cipher text "LYFORVEPRNHR" using key word "LEMON".

c) Given the encrypted message "QNXEPKMAEBTU" using Vigenere cipher keyword "key". Find original message.

a) Key: LEMON

PT: ATTACK AT PLACE

CT: LYFORVEPRNHRMH

|             |    |    |    |    |    |    |   |    |    |    |    |   |    |
|-------------|----|----|----|----|----|----|---|----|----|----|----|---|----|
| <u>Key:</u> | 11 | 4  | 12 | 14 | 13 | 11 | 4 | 12 | 14 | 13 | 11 | 4 | 12 |
| <u>PT:</u>  | 0  | 19 | 19 | 0  | 2  | 10 | 0 | 19 | 15 | 11 | 0  | 2 | 4  |
| <u>CT:</u>  | 11 | 23 | 5  | 14 | 15 | 21 | 4 | 5  | 17 | 24 | 11 | 6 | 16 |



crypt: LXFOPVBFRRHR

key: LEMON

|     |    |    |    |    |    |    |   |    |    |    |    |    |
|-----|----|----|----|----|----|----|---|----|----|----|----|----|
| Key | 11 | 4  | 12 | 14 | 13 | 11 | 4 | 12 | 14 | 13 | 11 | 4  |
| CT  | 11 | 23 | 5  | 14 | 15 | 21 | 4 | 5  | 17 | 13 | 7  | 17 |
| PT  | 0  | 19 | 19 | 0  | 2  | 10 | 0 | 19 | 3  | 0  | 22 | 13 |

PT: ATTACK AT DAWN

c) Cipher Text: QNXEPKMAEGTU

key: KEYKEYKEYKEY

PT: GOKAHACWAWPQ

Given a  $2 \times 2$  key matrix  $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

a) Encrypt the message "HI" using Hill cipher. (use  $A=0, B=1, \dots, Z=25$ )

b) If the message "PO" was encrypted to "QC" using the Hill cipher matrix, verify the result by showing the matrix multiplication.

Given matrix  $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

a) Encrypt "HI"

$H=7, I=8$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

$$45 \% 26 = 19 = T$$

$$54 \% 26 = 2 = C$$

Cipher text = TC

b) verify  $P \rightarrow QC$

$P=15, O=14$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

$$87 \% 26 = 9 = J, \quad 100 \% 26 = 22 = W$$



Actual result: JW

∴ PO → Qc is not correct for the Hill cipher

- a) Encrypt the message "MEET ME AFTER LUNCH" using key word "ZEBRAS"
- b) Decrypt the message "EANTTEFCUHE TREL" using key word "BOARDS"

a) PT: MEET ME AFTER LUNCH

Key: ZEBRAS

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| Z | E | B | R | A | S |
| 6 | 3 | 2 | 4 | 1 | 5 |
| M | E | E | T | M | E |
| A | F | T | E | R | L |
| U | N | C | H | X | X |

CT: MRXETCEFNTEHELXMAU

b) PT: EANTTEFCUHE TREL

Key: BOARDS

|   |   |   |   |   |
|---|---|---|---|---|
| B | O | A | R | D |
| 2 | 4 | 1 | 5 | 3 |
| M | H | E | R | F |
| T | E | A | E | C |
| E | T | N | L | U |

columns = 5  
 cipher text = 15  
 Rows =  $15 \div 5 = 3$

MHERF → 1, TEAEC → 2, ETNLU → 3

PT: MHERFTEAECETNLU

- a) using the one-time pad key XMCKL, encrypt the message "HELLO".
- b) Explain why the message "Attack Now" encrypted using OTP with key "QWERTYUIO" is unbreakable under ideal conditions.

a) PT: HELLO  
 Key: XMCKL  
 H → 7, E → 4, L → 11, L → 11, O → 14  
 X → 23, M → 12, C → 2, K → 10, L → 11

| Plaintext | Key | Addition | % 26 | Cipher letter |
|-----------|-----|----------|------|---------------|
| 7         | 23  | 30       | 4    | E             |
| 4         | 12  | 16       | 16   | Q             |
| 11        | 2   | 13       | 13   | N             |
| 11        | 10  | 21       | 21   | V             |
| 14        | 11  | 25       | 25   | Z             |



Ext  $\in QNVZ$

A One-Time Pad is a Symmetric encryption technique

→ The key is completely random.

→ The key length is equal to the message length.

→ Each key is used only once.

PT: ATTACKNOW      length = 9

Key: QWERTYUIO, length = 9

The same key is never reused. If the same key is reused, a hacker can compare ciphertexts and break the encryption.

Every possible plaintext is equally likely.

Encrypt the message

PT: "WE ARE DISCOVERED"      Key: "ZEBRA"

→ Arrange the msg in rows under the keyword

→ Determine column order based on alphabetical sorting of keyword letters

→ Read column in sorted order to form ciphertext

b) Decrypt the message

CT: EVLNEACD+KESEAGROFOSJDEECUXWN"

Key: TREAL

Task: Reconstruct the matrix based on the keyword.

→ Rearrange the column to the original order

→ Read row-wise to get the plaintext.

c) The message "INFORMATIONSECURITY" is encrypted using columnar transposition with keyword "HACK".

1) Encrypt the message

2) Explain how many rows are needed.



A) PT: WE ARE DISCOVERED

Key: ZEBRA

|   |   |   |   |   |
|---|---|---|---|---|
| Z | E | B | R | A |
| S | 3 | 2 | 4 | 1 |
| W | E | A | R | E |
| O | I | S | C | O |
| V | E | R | E | D |

EOD → 1  
 ASR → 2  
 EIE → 3  
 RCE → 4  
 WOV → 5

CT: EODASREIERCEWOV

b) CT: EVLNEACDTKESE QROFJDEE CUXWN

Key: TRIAL

length = 28

columns = 5

rows = 6

|   |   |   |   |   |
|---|---|---|---|---|
| T | R | I | A | L |
| S | 4 | 2 | 1 | 3 |
| W | E | A | R | E |
| O | I | S | C | O |
| V | E | R | E | D |
| F | L | E | E | A |
| T | O | N | C | E |
| X | X |   |   |   |

PT: WE ARE DISCOVERED FLEE AT ON CE

c) PT: INFORMATIONSECURITY

length = 14

Key: HACK

order: ACHK

|   |   |   |   |
|---|---|---|---|
| H | A | C | K |
| I | N | F | O |
| R | M | A | T |
| Z | O | N | S |
| E | C | U | R |
| I | T | Y | X |

CT: NHOC TFANUYIRIEZ OTSRX



given the following cipher text encrypted using a monoalphabetic substitution cipher: UZASOVUOHXMOVUGPOZPEVSGZWSZOPFPESEXU  
 E T U C G Assume it is an English sentence.

TASK:

- 1) perform frequency analysis of letters
- 2) use standard English letter frequencies (eg: E, T, A, O are most common) to hypothesize probable mappings.

3) Attempt to partially decrypt msg.

4) Provide the best guess for the plaintext message.

| letter | count | letter | count |
|--------|-------|--------|-------|
| U      | 6     | X      | 2     |
| Z      | 6     | H      | 2     |
| V      | 5     | M      | 1     |
| P      | 5     | W      | 1     |
| S      | 5     | F      | 1     |
| O      | 3     | C      | 1     |
| G      | 3     | Q      | 1     |
| E      | 3     |        |       |

The most frequent letters  $\rightarrow O=6, Z=6, U=5, P=5, S=5$

5)  $E \rightarrow T \rightarrow A \rightarrow O \rightarrow I$

| C | P |
|---|---|
| O | E |
| Z | T |
| V | A |
| P | O |
| S | I |

c) cipher: UZASOVUOHXMOVUGPOZPEE  
 VSGZWSZOPFPESEXUZE TUCG

After:

ATQIEVAEHXMEOVGOETOVZGT  
 WITBOFOISXATASTACH

more substitutions are required.

These are only probable mappings



d) since only frequency analysis is given, the exact plaintext cannot be uniquely found.

Final answer:

→ Frequency analysis completed

→ possible mappings found.

→ Partial decryption attempted.

→ Exact plaintext cannot be determined without additional clues.

9) you intercepted a message encrypted with a monoalphabetic substitution cipher:

CT: BJJW QJW YM JW NX FYNRQJWY

You know the original plaintext starts with: "THIS IS THE".

Task:

- 1) Use the known - plaintext position to find substitution mappings.
- 2) Complete the decryption of the full message.
- 3) validate if any letters repeat and adjust your substitution table if necessary.

2 CT: BJJW QJW YM JW NX FYNRQJWY

PT: THIS IS THE

1) BJJW QJW YM JW  
THIS IS THE

2) PT:

CT:

Reverse

cipher

Plain

|   |   |   |   |
|---|---|---|---|
| T | B | B | T |
| H | J | J | H |
| I | W | W | I |
| S | Q | Q | S |
| E | M | M | E |



$C + K = P$   
 53W QJW YM JW NY F XNRUQJWY

$P - K = C$   
 THIS IS THE \_\_\_\_\_

53W → THIS

QJW → IS

YM JW → THE

You are given the cipher text below, encrypted using the Vigenere cipher with an unknown short key (2-5 letter strong).

CT: ZICVTW QN GRZGV TWAVZH CQYGLH GJ

CT: ZICVTW AN GRZGV TWAVZH CQYGLH GJ

Encrypted with Vigenere cipher.

Key length: 2-5 letters

1) This is a Vigenere cipher, a polyalphabetic cipher

2) Find key: Kasiski Examination, Index of Coincidence (IC)

From the question, key length is already 2-5 letters

3) Recover the key: Split the ciphertext into groups according to the key length

4) Decrypt:  $P = (C - K + 26) \% 26$

where  $P$  = plaintext

$C$  = ciphertext

$K$  = key.

Since only the ciphertext is provided, the exact key and plaintext cannot be uniquely determined.